

CHAPTER IV

RESULTS AND DISCUSSION

4.1 Prototype Design and Implementation

This section answers Problem 1.

4.2 Prototype Performance under Normal Operating Conditions

This answers Problem 2.

4.2.1 Packet-Delivery Rate

4.2.2 End-to-End Latency

4.2.3 Transaction Success Rate

4.2.4 System Throughput

4.2.5 Raspberry Pi CPU and Memory Utilization

4.3 Security and Resilience Evaluation

This answers Problem 3.

4.3.1 Authentication and Access-Control Test

This section presents the results of the authentication and access-control tests conducted at the LoRaWAN, MQTT, and Hyperledger Fabric layers. The tests examined whether authorized devices and identities were correctly accepted and whether invalid credentials and prohibited actions were rejected. The results were assessed using the correct-decision rate, false acceptances, false rejections, response time, and unauthorized state changes. Table 12 summarizes the results of the 90 test attempts.

Table 12. Results of the Authentication and Access-Control Tests

Test layer	Test condition	Trials	Expected decision	Allowed	Rejected	Correct decisions, n (%)	False acceptance	False rejection
LoRaWAN	Registered device with correct OTAA credentials	10	Allow				—	
	Registered DevEUI with incorrect AppKey	10	Reject					—
	Unregistered DevEUI	10	Reject					—
MQTT	Authorized account publishes to an allowed topic	10	Allow				—	
	Account uses an incorrect password	10	Reject					—
	Limited account publishes to a prohibited topic	10	Reject					—
Hyperledger	Authorized writer submits a	10	Allow				—	

Fabric	sensor hash							
	Identity without writer permission submits a hash	10	Reject					—
	Invalid or untrusted identity submits a transaction	10	Reject					—

Note. False acceptance refers to an unauthorized request that was incorrectly allowed. False rejection refers to an authorized request that was incorrectly denied. An unauthorized state change occurs when a prohibited request successfully creates, modifies, or commits protected data. The dash indicates that the measure is not applicable to the test condition.

4.3.2 Replay and Spoofing Resistance Test

This section presents the results of the replay and spoofing tests conducted at the LoRaWAN layer. The assessment examined whether legitimate uplink messages were processed normally and whether previously accepted or forged packets were rejected before reaching the application, database, and blockchain layers. The results were evaluated using the number of accepted and rejected packets, correct decisions, false acceptances, false rejections, unauthorized propagation, and decision time. Table 13 summarizes the results of the 40 test attempts.

Table 13. Results of the Replay and Spoofing Tests

Test	Test condition	Trials	Expected decision	Received by gateway	Accepted by ChirpS	Rejected by Chirp	Reached MQTT	Database record	Fabric transactions	Correct decision	False accept	False reject	Mean decision time
------	----------------	--------	-------------------	---------------------	--------------------	-------------------	--------------	-----------------	---------------------	------------------	--------------	--------------	--------------------

			on	y, n	tack, n	Stack, n	/Node- RED, n	s create d, n	create d, n	ons, n (%)	ance, n	on, n	± SD (s)
Repla y	New legitim ate uplink	10	Accep t								—		
	Previo usly accept ed uplink with an old frame counte r	10	Reject									—	
Spoofi ng	Genui ne uplink from the registe red sensor	10	Accep t								—		
	Forged uplink using the legitim ate device addres s but an invalid MIC	10	Reject									—	
Overal l	All conditi ons	40	—										—

Note. False acceptance refers to a replayed or forged packet that was incorrectly accepted. False rejection refers to a legitimate packet that was incorrectly rejected. Unauthorized propagation occurs when an

attack packet reaches MQTT, Node-RED, PostgreSQL/TimescaleDB, or Hyperledger Fabric. Attack attempts not received by the gateway should be repeated and should not be counted as security decisions. The dash indicates that the measure is not applicable to the condition.

4.3.3 Data-Integrity Test

This section presents the results of the data integrity assessment conducted at the application and storage layers. The tests examined whether unchanged sensor records retained matching SHA-256 hashes and whether deliberately altered records were detected before valid storage or during comparison with the original hash recorded in Hyperledger Fabric. The results were evaluated using hash matches and mismatches, correct integrity decisions, false positives, false negatives, unauthorized storage, and verification time. Table 14 summarizes the results of the 40 data integrity test attempts.

Table 14. Results of the Data Integrity Tests

Test area	Test condition	Trial s	Expected result	Hash match, n	Hash mismatch, n	Correct decision s, n (%)	False positive s, n	False negatives, n	Unauthorized storage, n	Verification time, mean $\pm$ SD (s)
Application-layer integrity	Record processed without alteration	10	Hashes match, and record is stored					—	—	
Application-layer integrity	Sensor value altered before storage	10	Hash mismatch; record is blocked or quarantined				—			
Post-	Databas	10	Database					—	—	

storage integrity	e record remains unchanged		hash matches the Fabric hash							
Post-storage integrity	Database value altered after Fabric commitment	10	Recomputed hash differs from the Fabric hash				—		—	
Overall	All test conditions	40	—							—

Note. A false positive occurs when an unchanged record is incorrectly identified as altered. A false negative occurs when an altered record is incorrectly reported as valid. Unauthorized storage refers to an application-layer altered record that is accepted and stored as a valid operational record. A dash indicates that the measure is not applicable to the condition.

4.3.4 Traceability Test

This section presents the results of the traceability assessment. The tests examined whether individual sensor readings and groups of consecutive readings could be completely retrieved and correctly linked across PostgreSQL/TimescaleDB and Hyperledger Fabric. Table 15 summarizes the retrieval, completeness, linkage, chronological accuracy, and time required for the traceability tests.

Table 15. Results of the Traceability Tests

Traceability test	Trials	Records tested	Successfully retrieved, n (%)	Complete records, n (%)	Correct database – blockchain	Correct chronological order, n	Missing records, n	Duplicate records, n	Mean retrieval time $\pm$ SD (s)
-------------------	--------	----------------	-------------------------------	-------------------------	-------------------------------	--------------------------------	--------------------	----------------------	----------------------------------

					ain links, n (%)	(%)			
Individu al sensor- record trace	10	10				Not applicab le			
Device- history reconstr uction	10 sequenc es	50							
Overall	20 trials	60							—

4.3.5 DoS and Flooding Resistance Test

This section presents the results of the MQTT connection-flooding and invalid application-message flooding tests under normal, moderate, and high traffic conditions. The results were assessed using legitimate-message delivery, invalid-traffic rejection, end-to-end latency, resource utilization, service availability, unauthorized records, and recovery time. Table 16 summarizes the results of the 18 experimental runs.

Table 16. Results of the DoS or Flooding Tests

Test area	Traffic condition	Valid messages delivered, n (%)	Invalid traffic rejected, n (%)	Mean latency $\pm$ SD (s)	Mean CPU use (%)	Mean memory use (%)	Service available	Unauthorized records, n	Mean recovery time $\pm$ SD (s)
MQTT connection flooding	Normal		Not applicable						Not applicable

	Moderate								
	High								
Invalid application-message flooding	Normal		Not applicable						Not applicable
	Moderate								
	High								

Note. Each condition consists of three five-minute runs. Approximately 60 legitimate sensor readings are expected for each condition. Moderate and high flooding generate approximately 9,000 and 45,000 invalid requests, respectively, across the three runs. Service availability should be reported as Available or Unavailable.

4.3.6 Resilience and Recovery Test

This section presents the results of the Internet-connectivity interruption and recovery test. The assessment examined whether the prototype continued collecting, processing, storing, displaying, and recording sensor data while Internet access was unavailable and whether normal connectivity was restored without missing, duplicate, or incorrectly linked records. Table 17 summarizes the results from the three experimental runs.

Table 17. Results of the Internet-Connectivity Interruption and Recovery Test

Test period	Total duration across three runs	Expected readings	Readings stored in database, n (%)	Fabric records created, n (%)	Missing records, n	Duplicate records, n	Local services available	Mean end-to-end latency $\pm$ SD (s)	Mean recovery time $\pm$ SD (s)
Normal operation before interruption	90 minutes	360							Not applicable
Internet interruption	180 minutes	720							Not applicable

Recovery after reconnection	90 minutes	360							
Overall	360 minutes	1,440					—	—	—

Note. Local services include ChirpStack, Mosquitto MQTT, Node-RED, PostgreSQL/TimescaleDB, Grafana, and Hyperledger Fabric. Service availability should be reported as Available, Partially Available, or Unavailable. Recovery time refers to the period from restoration of Internet connectivity until external connectivity and related services return to normal operation.